

LECTURE 07

Natural Language Processing

Regularization & Gradient-Based Optimization
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CONTROLLING OVERFITTING AND LEARNING BETTER PARAMETERS



Today's Objective Is Better Generalization

STAGE 01

Model Learns

Estimating parameters from training data using Logistic Regression.

STAGE 02

Model Overfits

Memorizing training noise instead of learning reusable patterns.

STAGE 03

Regularization

Controlling complexity to ensure the model stays simple.

STAGE 04

Optimization

Using Gradient Descent to find the best parameter values.

THE GOAL

Move from making simple predictions to learning parameters that generalize to unseen data.

NLP Models Evolve from Features to Optimization

LECTURE 03

Text Representation

Text → Numerical Features (BoW / TF-IDF)

LECTURE 04

Naïve Bayes

Probabilistic classification by counting occurrences

LECTURE 05

Perceptron

Linear discriminative learning through updates

LECTURE 06

Logistic Regression

Adding Probabilities and Log Loss functions

TODAY

Regularization

Optimization for better generalization and stability

BUILDING THE FOUNDATION FOR ROBUST LANGUAGE MODELS

Learning Means Estimating Parameters

DATA
TRAINING SET



ALGORITHM
OPTIMIZATION



MODEL
PARAMETERS



OUTPUT
PREDICTIONS

LOGISTIC REGRESSION
PARAMETERS

$w_1, w_2, \dots, w_n, b$

The model searches for parameter values that minimize prediction error. Each weight (w) determines how strongly a specific feature influences the final classification score.

Training Success Must Transfer to Unseen Text

KNOWN DATA

Training Set

The model observes these examples to estimate its internal parameters (weights and bias).

UNKNOWN DATA

Test Set

The true measure of utility. Performance here indicates how well the model generalizes.

Training Performance $\approx$ Test Performance

A LARGE PERFORMANCE GAP INDICATES A MODEL-COMPLEXITY PROBLEM.

Underfitting Means the Model Is Too Simple

ERROR METRICS

Training Error **HIGH**

Test Error **HIGH**

The model fails to capture the underlying structure of the data because it lacks sufficient complexity.

NLP EXAMPLE

Trying to classify sentiment (Positive/Negative) using only a single feature:

Sentence Length

Length alone cannot distinguish between "This was a long, boring movie" and "This was a long, fantastic movie."

Underfitting occurs when the model is unable to learn the true relationship between features and labels.

Overfitting Means Memorizing Noise

TRAINING DATA

Very Low

MEMORIZATION

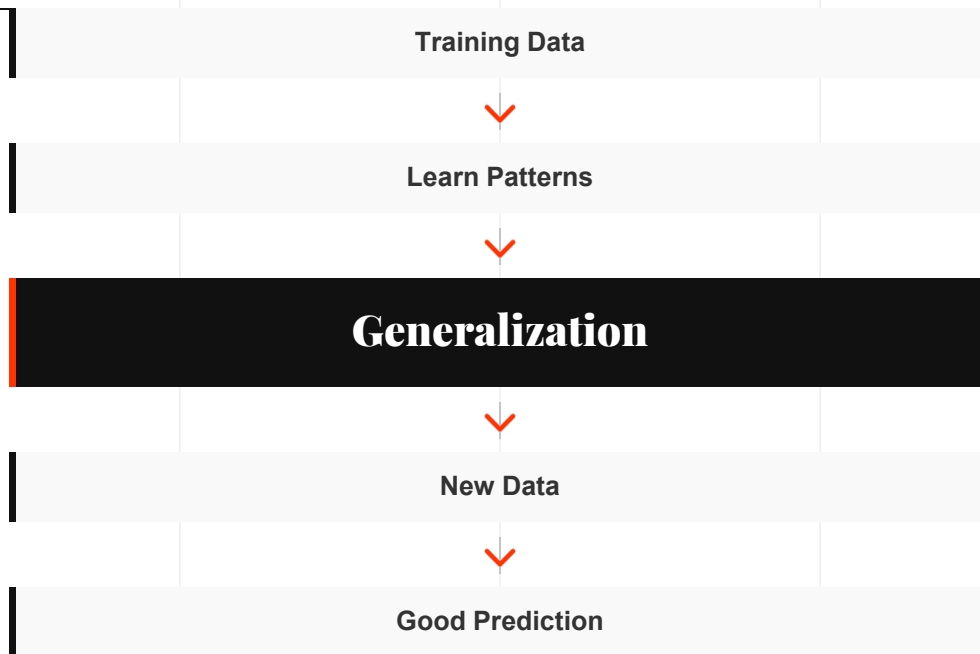
TEST DATA (UNSEEN)

High

FAILURE TO GENERALIZE

The model learns useful patterns, noise, and accidental correlations. **IT MEMORIZES INSTEAD OF GENERALIZING.**

Generalization Is the Real Learning Objective



THE CORE PRINCIPLE: A model should perform well on examples it has never seen. The goal is not merely to minimize loss, but to ensure the model works in the real world.

NLP Overfitting Can Attach Labels to Names

USEFUL PATTERN

"The movie was absolutely fantastic."

fantastic → positive

GENERALIZES WELL

ACCIDENTAL CORRELATION

"I loved watching XYZMovie today."

XYZMovie → positive

OVERFITTING RISK

Overfitting occurs when a model learns specific names or noise that appear predictive in training but fail to generalize to new, unseen data.

Regularization Adds a Complexity Penalty

THE REGULARIZED OBJECTIVE

Minimize: Prediction Loss + Complexity Penalty

PREDICTION LOSS

Rewards the model for making accurate predictions on the training data (e.g., Log Loss).

COMPLEXITY PENALTY

Discourages extreme or large weights, pushing the model toward simpler, more stable patterns.

REGULARIZATION ENSURES THE MODEL STAYS USEFUL WITHOUT BEING OVERLY SPECIFIC.

The Regularized Objective Balances Fit and Simplicity

LOGISTIC REGRESSION OBJECTIVE

$$J(w) = \text{Log Loss} + \lambda \times \text{Penalty}$$

Data Loss

Rewards accurate predictions on the training set. (Log Loss)

$\lambda \times \text{Penalty}$

Discourages overly complex parameter settings (large weights).

The regularization strength λ controls the trade-off between fitting the data and keeping the model simple.

Increasing λ Produces a Simpler Model

NO REGULARIZATION

$\lambda = 0$

Weights are unconstrained by a penalty. High
High risk of overfitting.

WEAK REGULARIZATION

Small λ

Modest control of large weights. Balanced
complexity.

STRONG REGULARIZATION

Large λ

Aggressive weight shrinkage. Produces a very
very simple model.

INCREASING λ



SMALLER WEIGHTS



SIMPLER MODEL



LESS OVERFITTING

L2 Regularization Penalizes Large Weights

THE L2 PENALTY TERM

$$\text{Penalty} = \lambda \sum w_i^2$$

L2 regularization adds the **squared magnitude** of the weights to the objective function. Because large weights are squared, extreme feature effects receive disproportionately larger penalties, encouraging the model to favor **small, distributed weights**.

Squaring Makes Large Weights Expensive

BASE WEIGHT

$$w_1 = 1$$

1

RELATIVE PENALTY (1^2)

VS

LARGE WEIGHT

$$w_2 = 5$$

25

RELATIVE PENALTY (5^2)

A weight 5x larger creates a **25x larger** penalty. L2 encourages small, distributed weights rather than a few extreme feature effects.

L2 Makes Feature Influence Less Extreme

UNREGULARIZED WEIGHTS

w_1 8.2

w_2 -7.4

w_3 6.9

Overly sensitive to individual features.



SHRINKAGE



L2 REGULARIZED WEIGHTS

w_1 1.8

w_2 -1.4

w_3 1.6

Stable, distributed influence.

L2 regularization produces less extreme weights, making the model more robust and less likely to overfit to specific training examples.

L1 Regularization Can Create Exact Zeros

THE L1 PENALTY TERM

$$\text{Penalty} = \lambda \sum |w_i|$$

L1 regularization uses the **absolute values** of the weights. Unlike L2, which shrinks weights toward zero, L1 has a unique mathematical property that can drive some weights **exactly to 0**. This leads to **sparse models** where only a subset of features are active.

L1 Performs Feature Selection

RETAINED SIGNALS

excellent	2.4
fantastic	1.9
wonderful	2.1

Predictive features kept.

IRRELEVANT FEATURES

movie	0
the	0
is	0

Noise and stop words removed.

With 10,000+ words, L1 identifies the most useful signals. **ZERO-WEIGHT FEATURES** effectively become irrelevant to the model's classification logic.

L1 and L2 Make Different Trade-Offs

PROPERTY	L1 REGULARIZATION	L2 REGULARIZATION
Penalty Type	Absolute Values ($\sum w $)	Squared Values ($\sum w^2$)
Zero Weights	Common (Sparsity)	Rare (Shrinkage)
Feature Selection	✓Yes	Usually No
Weight Shrinkage	✓Yes	✓Yes
Sparse Features	Useful	Useful

Choose L1 for feature selection and interpretability, or L2 for stability and handling correlated features.

Elastic Net Combines Sparsity and Stability

THE ELASTIC NET PENALTY

$$\text{Penalty} = \alpha \sum |w_i| + (1-\alpha) \sum w_i^2$$

L1 Contribution

Provides feature selection by driving irrelevant weights to exactly zero.

L2 Contribution

Provides weight stabilization, preventing extreme sensitivity to correlated features.

Highly effective when features are numerous and correlated, offering the best of both worlds.

Optimization Finds Better Parameters

THE CHALLENGE

NLP models like Logistic Regression often contain thousands or millions of parameters that must be estimated simultaneously.

Find: $w_1, w_2, \dots, w_n$ that minimize $J(w)$

Training is an optimization problem over a high-dimensional loss surface. We need an efficient way to navigate this surface to find the global minimum.

Gradient Descent

The central method for iteratively improving parameters by moving in the direction of steepest descent.

Gradient Descent Moves Downhill

THE MOUNTAIN METAPHOR

"Imagine standing on a mountain in thick fog. Your goal is to reach the lowest point in the valley by only feeling the slope beneath your feet."

STEP 01

Determine the Slope

STEP 02

Move Downhill

STEP 03

Repeat Until Flat

Machine learning applies this logic to the **LOSS FUNCTION** to find the best parameters.

The Gradient Points Toward Increasing Loss

The gradient identifies the direction in which loss increases most rapidly. To reduce loss, we update parameters in the **opposite direction**.

THE PARAMETER UPDATE RULE

$$\theta_{\text{new}} = \theta_{\text{old}} - \eta \nabla J(\theta)$$

θ

MODEL PARAMETERS

η

LEARNING RATE

∇J

GRADIENT OF LOSS

The Learning Rate Controls Step Size

SMALL H

Too Small

Optimization is safe but **very slow**. The model requires many iterations to reach the minimum.

OPTIMAL H

Good

Produces **steady convergence**. The model reaches the minimum efficiently without instability.

LARGE H

Too Large

Causes **overshooting** and oscillation. The model may fail to converge or even diverge.

Learning rate selection determines whether optimization is efficient and stable.

A Gradient Update Moves the Weight Downhill

INITIAL CONDITIONS

Current Weight	2.0
Gradient	0.8
Learning Rate	0.1

UPDATE CALCULATION

$$w_{\text{new}} = w_{\text{old}} - (\eta \times \nabla J)$$

$$w_{\text{new}} = 2.0 - (0.1 \times 0.8)$$

NEW WEIGHT VALUE

The weight moves in the direction that reduces the loss, effectively descending the error surface.

Gradient Descent Repeats a Structured Training Loop

STEP 01 Initialize Parameters (w , b)



STEP 02 Calculate Predictions



STEP 03 Calculate Loss (Log Loss + Penalty)



STEP 04 Calculate Gradient



STEP 05 Update Parameters

The loop continues until the model reaches an acceptable solution.

Batch Gradient Descent Uses Every Training Example

DATA

ENTIRE CORPUS



GRADIENT

AVERAGE OF ALL



UPDATE

SINGLE STEP

ADVANTAGES

Stable gradients and smooth learning curves. Guarantees convergence to the global minimum for convex problems.

DISADVANTAGES

Computationally expensive for very large datasets. Requires loading the entire dataset into memory.

Stochastic Gradient Descent Trades Stability for Speed

EXAMPLE 1

Update

EXAMPLE 2

Update

EXAMPLE 3

Update

EXAMPLE 4

Update

EXAMPLE 5

Update

ADVANTAGES

Fast updates and extremely suitable for very large datasets. Can help the model escape local minima.

DISADVANTAGES

Noisy updates and fluctuating loss curves. May never perfectly settle at the global minimum.

Mini-Batches Balance Efficiency and Stability

BATCH SIZE

32

=

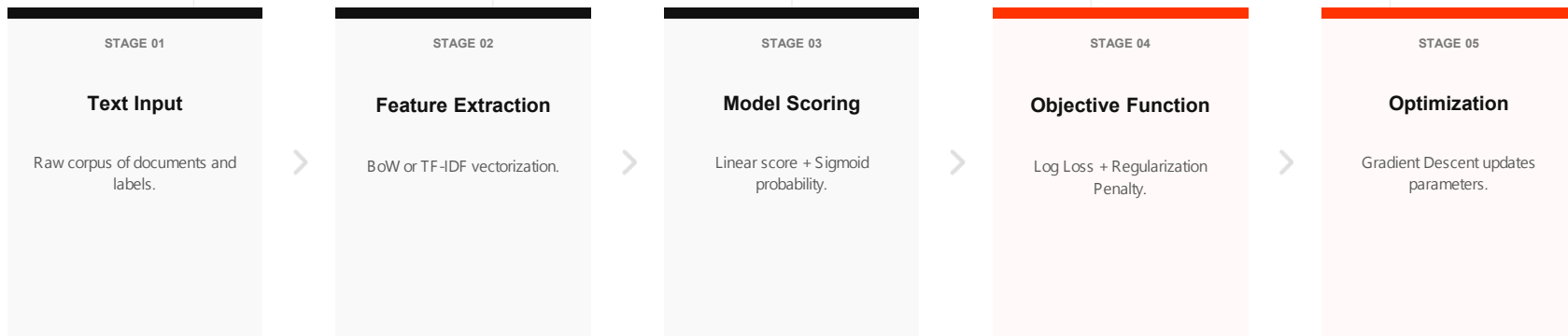
UPDATE FREQUENCY

Every 32 Examples

Mini-Batch Gradient Descent combines the **smoothness** of batch learning with the **speed** of stochastic updates. It is the industry standard for modern machine learning and deep learning.

The "Goldilocks" choice: Not too slow, not too noisy.

Regularized Logistic Regression Closes the Loop



The model computes a linear score, applies the sigmoid, and produces a probability. Training combines **prediction loss** with **regularization**, **regularization**, updating parameters until convergence.

Regularization and Optimization Improve Generalization

OVERFITTING CONTROL

Regularization prevents models from learning noise by penalizing complexity.

L1 VS L2

L1 creates sparsity and feature selection; L2 shrinks weights for stability.

OPTIMIZATION

Gradient Descent iteratively navigates the loss surface to find optimal parameters.

HYPERPARAMETERS

Learning rate and update strategy control the speed and stability of training.

NEXT LECTURE

Model Evaluation Metrics

Precision, Recall, F1-Score